Shrestha Bhattacharya

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SUMMARY

Data engineer with 3+ years of experience building, optimizing, and supporting SQL data pipelines and validating data quality for brokerage and insurance claims data, completing an M.S. in Analytics at Georgia Tech (GPA 3.83/4.00). Cut one pipeline's runtime by ~87.5% and traced recurring production failures to their source. Hands-on Python (pandas, scikit-learn) for statistical modeling and machine learning, with foundational knowledge of A/B testing, hypothesis testing, causal inference, and product metrics. Uses Claude for coding, debugging, and data exploration, verifying every output. Most at home when a number looks wrong: tracing it back to its source, then explaining the cause and the fix to the people who have to act on it, technical or not.

TECHNICAL SKILLS

- **Data Engineering:** Data Pipelines, ETL, Data Modeling (Slowly Changing Dimensions Type 1 & 2), Data Validation, SQL Optimization, Informatica PowerCenter, AQT, Autosys
- **Languages & Libraries:** Python (pandas, NumPy, scikit-learn, Matplotlib, NLTK), SQL (PostgreSQL)
- **Statistics & Experimentation:** Experimental Design, A/B Testing, Hypothesis Testing, Regression Modeling, Causal Inference, Forecasting, Predictive Modeling
- **AI Tools:** Claude for coding, debugging, and data exploration

PROFESSIONAL EXPERIENCE

Cognizant Technology Solutions, India — Jul 2021 - Nov 2024

Associate

- Reduced data pipeline runtime by approximately 87.5% by optimizing SQL transformations: filtering records earlier and removing unnecessary joins.
- Built and supported SQL data pipelines using Slowly Changing Dimension Type 1 and Type 2 logic for brokerage and insurance claims data, producing reliable datasets for downstream reporting.
- Redesigned a problematic data load with an intermediate table refreshed weekly, improving control over source-data issues and reducing downstream failures.
- Monitored scheduled workflows and resolved production data issues under time-sensitive requirements; traced recurring failures to client source-file errors, worked with the client to correct inputs, and recommended automating the recurring preparation step.
- Partnered with cross-functional technical and client teams to gather requirements and translate business rules into production transformations; supported deployment, testing, and post-release validation of pipeline changes.

PROJECTS

YouTube Trending Content Analytics | *Georgia Tech Course Project*

- Compared several statistical learning models to predict log-transformed views on 56,640 unique videos deduplicated from a 161,470-record, high-dimensional trending dataset, preventing train/test leakage.
- Random Forest achieved test RMSE 1.41 vs. a 1.75 mean baseline (~19% reduction, $R^2 \approx 0.35$), with stable cross-validation (CV RMSE 1.43, SD 0.004); residual analysis showed error concentrated in viral-video outliers.
- Used Random Forest feature importance and LASSO selection to identify video category, title length, tag count, and publishing country as the strongest predictors of audience behavior, and built a separate model to quantify how much post-publication engagement signal adds.

Movie Rating Analysis & Recommendation System | *Python, pandas, scikit-learn*

- Built a content-based movie recommender on the TMDB 5000 dataset using TF-IDF on combined overview, genre, and keyword text with cosine similarity.
- Implemented user-based collaborative filtering on MovieLens 100K from a user-item rating matrix to recommend highly rated movies a user has not yet seen.
- Trained a TF-IDF + LinearSVC sentiment classifier on IMDB reviews, reaching 81.75% accuracy and 0.90 AUC on a held-out test set.

EDUCATION

Georgia Institute of Technology, Atlanta, GA — Aug 2025 - Aug 2027 (Expected)

M.S. in Analytics, Computational Data Analytics Track | GPA: 3.83/4.00

University of Engineering & Management, Kolkata, India — Aug 2017 - May 2021

B.Tech in Computer Science & Engineering | CGPA: 8.58/10.00